MAT3032 Advanced Algebra Unassessed Coursework 2

Please upload your answers to the assignment folder in SurreyLearn by 4:00 pm on Friday 6th December Week 11 2024. There are some hints on the next page, but please attempt the questions first. You may wish to leave Questions 4, 5 and 6 until after we cover Section 2.4, 2.5 and 3.1 respectively.

1. Let H be an *abelian* proper subgroup of a multiplicative group G with identity element e. For any $a, b \in H$ and $x \in G$, define $(a, b)(x) = axb$.

   (a) Show that this defines a group action of $H \times H$ on G.

   (b) Identify the orbit and stabiliser of e and the fixed set of (e, e).

2. A group G, of order 12, has presentation $\langle x, y \ : \ x^4 = y^3 = e, \ yxy = x \rangle$ where e is the identity element. Thus $G = \{x^r y^s \ : \ r \in \{0, 1, 2, 3\}, \ s \in \{0, 1, 2\}\}$.

   (a) Show that $x^2 y = y x^2$. Deduce that $x^2 \in Z(G)$.

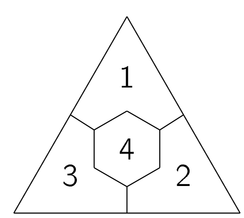
   (b) You are given that each of the following is a conjugacy class in G: $\{x, xy, xy^2\}$, $\{x^3, x^3 y, x^3 y^2\}$, $\{x^2 y, x^2 y^2\}$. Find the other conjugacy classes.

   (c) Show that G has normal subgroups of orders 2 and 3.

   (d) Identify a group which is isomorphic to $G/Z(G)$.

3. Let σ be the permutation $(1 \quad 2)(3 \quad 4)(5 \quad 6 \quad 7)$ in the symmetric group S_7. State the order, cycle index and cycle type of σ. Find an expression (which need not be evaluated) for the number of elements in S_7 which have the same order as σ.

4. An equilateral triangle is divided into four regions as shown. The regions labelled 1, 2 and 3 are congruent.
   The figure is cut out of transparent material.



   The regions are to be coloured using q available colours, each of which may be used any number of times. Find the number of q-colourings of the figure if

   (a) it can only be rotated in its plane,

   (b) it can also be turned over.

5. Let G be a group of order $392 = 2^3 \times 7^2$.

   (a) Find the possible numbers of p-Sylow subgroups of G for $p = 2$ and $p = 7$. State the orders of these subgroups.

   (b) Show that G is not a simple group.

(c) Find, up to isomorphism, how many groups of order 392 have both a unique 2-Sylow and a unique 7-Sylow.

6. Use suitable quaternions to find $p, q, r, s \in \mathbb{N}$ such that $(1^2+3^2+5^2+7^2)(2^2+4^2+6^2+8^2) = p^2+q^2+r^2+s^2$.

Hints

1. (a) Check the three defining properties of a group action. Remember that in a direct product we have $(a, b)(c, d) = (ac, bd)$, and (e, e) is the identity.
   (b) The orbit and fixed set are contained in the set being acted on, namely G, while the stabiliser is a subgroup of the group that acts, $H \times H$.

2. (a) $x^2 y = x(xy) = \ldots$ If x^2 commutes with the generators of G then it must commute with all elements.
   (b) Each element of the centre is in a one-element conjugacy class. Three other classes are given. Are the remaining elements in the same class?
   (c) Can you find two subgroups which are unions of conjugacy classes? (Proposition 2.12.)
   (d) Which groups of order $|G/Z(G)|$ exist? Which of these is not possible, as in 1(b)?

3. The order is the lcm of the cycle lengths. Which other cycle types in S_7 have the same order? Use Proposition 2.16.

4. List (a) the rotational symmetries, (b) all the symmetries of the figure, as permutations of $\{1, 2, 3, 4\}$.

   Use Proposition 2.18.

5. (a) You should find three possible values for w_2 and two for w_7.
   (b) Show that if G is simple then it is isomorphic to a subgroup of S_8, which is impossible - why?
   (c) Apply Proposition 2.22 (iii). Which groups of orders 8 and 49 exist?

6. Consider Proposition 3.2.